

Younghan Park

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Research Interests

Multimodal Machine Learning, Natural Language Processing, Grounded Language Understanding

Education

Carnegie Mellon University, Pittsburgh, PA, USA Aug 2026 – Dec 2027 (Expected)
MS in Machine Learning, School of Computer Science

Yonsei University, Seoul, South Korea Mar 2020 – Aug 2025
BS in Computer Science and Engineering, College of Computing GPA: 4.14/4.30 (98.8/100)

- **Honors** (Fall 2020, Spring 2021, Spring 2023), **High Honors** (Fall 2024)
- **Graduated with High Honors** (awarded to 2 out of 88 students in the College of Computing)
- Relevant Coursework: Computer Vision, Machine Learning, Big Data, Multimodal Deep Learning (Grad)
- Leave of Absence for Military Service (Jul 2021 - Jan 2023)

The University of Texas at Austin, Austin, TX, USA Aug 2023 – May 2024
Exchange Student (Major: Computer Science) GPA: 4.00/4.00

- **University Honors** (Fall 2023)
- Relevant Coursework: Natural Language Processing, Spoken Language Technologies (Grad), Reinforcement Learning: Theory & Practice (Grad)

Honors and Awards

The Korean Government Scholarship Program for Study Overseas Starting Fall 2026
◦ \$80,000 for two years (selected as one of three recipients in the *digital* category nationwide)

Korean Presidential Science Scholarship Mar 2024 – Aug 2025
◦ Full tuition and stipend for junior and senior years, awarded in the name of the President of the Republic of Korea (selected as one of 60 students nationwide)

Merit-based Scholarship, Yonsei University Spring 2021, Spring 2023, Fall 2023

Best Tutor, Yonsei University Spring 2021

Runner-Up, KPMG Ideathon Challenge 2022 National Final, KPMG Korea Feb 2022

3rd Place, KPMG Ideathon Challenge 2021 National Final, KPMG Korea Feb 2021

Publications and Patents

*: equal contribution, †: equal advising

Preprints / Under Review

[P1] Minju Jeon, **Younghan Park**, Han Sung Park, Jong-Hwan Kim, Dong-Jin Kim, Hoyeon Lee. "Phonemizing User-Generated Text: A Benchmark, Taxonomy, and Compositional Approach". Under Review at *Empirical Methods in Natural Language Processing (EMNLP)* 2026

Conference Proceedings / Workshop Papers

[C2] **Younghan Park***, Hoyeon Lee*, Hawon Jeong, Jong-Hwan Kim. "LLM-Based Multi-Reference Evaluation for Efficient and Robust Assessment of Phrase Break Annotations". *Interspeech* 2026

[W1] Seoyeon An*, Hyeonseo Jang*, Minsu Kim*, Chanho Lee, **Younghan Park**, Kangwook Lee. "AgentVid-Bench: A Multi-Hop Video Question Answering Benchmark for Evaluating MLLM Agents". *Workshop on Failure Modes of Agentic AI @ International Conference on Machine Learning (FAGEN@ICML)* 2026

[C1] **Younghan Park**, Anuj Diwan, David Harwath[†], Eunsol Choi[†]. "Rhapsody: A Dataset for Highlight Detection in Podcasts". *Conference on Language Modeling (COLM)* 2025

Patents

[Pa1] Kyuhyun Hwang*, Gideok Jang*, Woonchae Jeong*, Jaein Yoo*, **Younghan Park***, Yonsoo Kim*. "Control Apparatus Using Oral Cavity". Korea Patent Registration No. 10-2020-0140424. Granted.

Experience

KRAFTON Inc. (Publisher of PUBG: Battlegrounds), Seoul, South Korea
Intern, AI Applied Research Dept., AI Research Division

Dec 2025 – Present

- Built a two-stage PUBG anti-cheat pipeline: log-based localization → investigation with multimodal LLMs.
- Developed a log-based cheat localization model, treating game logs as irregular multivariate time series data and applying approaches from weakly-supervised temporal action localization literature.
- Designed an automatic replay video investigator, exploring training-free/inference-time methods, fine-tuning open-source LLMs (SFT via distillation, RLVR), and agentic approaches [W1].
- Evaluated how well PUBG's MMR predicts future match kills and placement, comparing it against existing rating systems as well as a new, data-driven rule learned from past matches.

NAVER Cloud Corp., Seongnam, South Korea
Research Intern, Voice Technology Team

Jun 2025 – Oct 2025

- Proposed an LLM-based multi-reference evaluation framework for phrase break annotations, improving Pearson correlation from 0.50 to 0.63, without the human involvement required by prior methods [C2].
- Improved a phrase break prediction model for TTS, reducing training time to under a third and inference latency by 67% with no performance loss. Optimized the model for deployment (PyTorch → ONNX).

NLP Group, The University of Texas at Austin

Aug 2023 – May 2025

Visiting Student (Advisors: Prof. [Eunsol Choi](#), Prof. [David Harwath](#))

- Built a scalable pipeline to construct a 13K-episode podcast highlight detection dataset and trained a text-audio model, improving F1 by +14%pt. over GPT-4o, using only 6.2M trainable parameters [C1].
- Converted a terabyte-scale audio corpus into a conversational TTS dataset. Split raw audio into dialogues, extracted prosody features, and extended FastSpeech2 with a conversation history encoder.

Multimodal Intelligence Research Lab, Yonsei University
Undergraduate Research Intern (Advisor: Prof. [Youngjae Yu](#))

Feb 2023 – Aug 2023

- Investigated approaches for training a video highlight detection model from large-scale web videos.

Korean Augmentation to the US Army (KATUSA), USAG Humphreys
Specialty: Military Police, Final Rank: Sergeant

Jul 2021 – Jan 2023

MouthMouse, Seoul, South Korea
Co-Founder, Software Engineer

Apr 2020 – Jul 2021

- Co-founded a startup and developed an oral human-computer interaction device, enabling easier smart device interaction for individuals with physical disabilities; patent granted [Pa1].
- Won 5+ startup competitions, secured ~\$30,000 in total prize money, and was featured in Korean media ([The Seoul Economic Daily](#), [The Digital Times](#), [The Korea Economic Daily](#)).

Selected Projects

RL Agent for Multi-turn Negotiation in Chat Conversations

Spring 2024

- Built a multi-turn negotiation agent by fine-tuning an LLM to propose multiple candidate utterances and buy prices per turn. Trained offline RL agents from model response representations [\[report\]](#).

Robust Speaker Diarization under Real-world Noise

Fall 2023

- Built a speaker diarization pipeline and evaluated 11 preprocessing configurations, including source separation and speech enhancement, under synthetic background noise and crosstalk [\[report\]](#).

Teaching Experience

Tutor, Yonsei CSI2100: Computer Programming

Spring 2021

Skills

Languages: Korean (Native), English (Fluent; TOEFL 108/120)

Programming: Python, Shell, C/C++, Web

Frameworks: PyTorch, HuggingFace (Transformers, Datasets, PEFT, TRL), unsloth, vLLM

Infra & Tools: Linux, Git, distributed training (DDP, FSDP), Spark, ONNX